ADA University, Fall 2023, Discrete Structures / Fundamentals of Math

SAMPLE TASKS for the Midterm 1 (October 14, 2023). DURATION: 90 Min

25 points

Full Name: _____ CRN: _____

- Special sequences for the case of THREE input variables A, B, C (or x, y , z):

A (or x) = 00001111, B (or y) = 00110011, C (or z) = 01010101

- The truth table for the minterms (**using the special sequences for x, y, z**) are as follows:

Table 1

Truth tables for minterms using the special sequences for x, y, z

xyz=00000001	xyz′=00000010	xy′z=00000100	xy′z′=00001000
x′yz=00010000	x′yz′=00100000	x′y′z=01000000	x′y′z′=10000000

SAMPLE Tasks for Midterm I

Task 1 (4 pt). Design a three-input **minimal** AND-OR circuit L with the following truth table:

T =[x, y, z; L]=[00001111, 00110011, 01010101; **01011111**]

Hint.
Step 1. Write truth table as a sum of minterms. (**1 pt**)
Step 2. Create a Karnaugh Map based on result of the Step 1. (**1pt**)
Step 3. Find an appropriate MSoP (**1 pt**)
Step 4. Design a three-input **minimal** AND-OR circuit L which corresponds to the Step 3

Task 2 (4=2+1+1 pt).

Find the truth table T=T(E) for the Boolean expression

E=E(x, y, z)=(x′y′+z)′yz′+(x′z+y)′(y′z+x)+x′z+xy′z′

Hint.
Step 1. Find SoP form for E. (**2 pt**)
Step 2. Find CSoP form (**1pt**)
Step 3. Use Table 1 to find T(E) (T(E) will be the sequence of eight digits, 0 or 1). (**1 pt**)

Task 3 (3 pt=1+2)

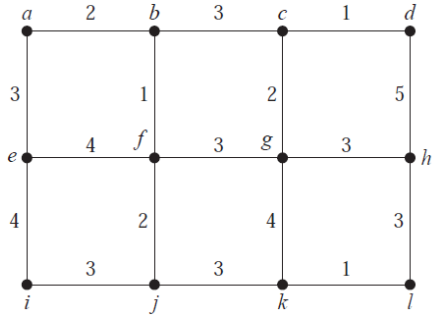
Find a Boolean expression E=E(x, y, z) in **MSoP form** corresponding to the truth table T(E):

T(E)=11011100;

Hint.

1. Find CSoP form for E by using Table 1. (**1 pt**)
2. Then use Karnaugh map method to reduce CSoP form to an MSoP form. If there exist two or more MSoP forms it will be enough to find only one MSoP form. (**2 pt**)

Task 4 (3 pt) Use Prim's algorithm to find a minimum spanning tree in the graph.



Task 5 (4=2+2 pt)

1. Find a MSoP for the Boolean expression E given by Table (A). (*Write as an expression*). **(2pt)**
2. Find a MSoP for the Boolean expression E given by Table (B). (*Write as an expression*). **(2pt)**

(A)	**zt**	**zt′**	**z′t′**	**z′t**
xy	V	V		V
xy′			V	V
x′y′	V	V		
x′y	V	V	V	V

(B)	**zt**	**zt′**	**z′t′**	**z′t**
xy	V			V
xy′		V	V	V
x′y′	V	V	V	V
x′y	V	V		

Task 6 (5 pt =1+2+2)

Subtask 6.1 (1 pt). Find the negation of the proposition below

$\forall x \exists y \forall z P(x, y, z)$

Subtask 6.2 (2 pt). Domain of all predicates P(x) below is the set of *all integers*. What are the truth values of the corresponding propositions?

a) $\exists !x(x>1)$ **(0.5 pt)**

b) $\exists !x(x^2=1)$ **(0.5 pt)**

c) $\exists !x(x+3=2x)$ **(0.5 pt)**

d) $\exists !x(x=x+1)$ **(0.5 pt)**

Good Luck!

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